Prakruth Adari

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Education

2021 - present Ph.D., Physics and Astronomy at **Stony Brook University**
2016 - 2020 B.A. in Physics and Math at **Columbia University**

Research Experience

LSST Dark Energy Science Collaboration (DESC)

Catalog based detection of unrecognized blends - Use of AI/ML tools to detect unrecognized blends at the catalog level. Proof of concept work that could be used to improve weak lensing.

Unrecognized blends + Anacal Shear Calibration - Testing how removal of unrecognized blends can help improve shear calibration within the Anacal framework. Publication under review.

DP2 Cluster - Leading efforts to use AnaCal shears to measure tangential shear profiles around clusters in Rubin Data Preview 2. This includes validation of shears and photometry via image simulation, estimating calibration biases, and testing source selection schemes.

Vera C. Rubin Observatory

Cluster commissioning - Stress testing Rubin pipelines by conducting cluster weak lensing science. Worked on source selection, Anacal shear measurements, and $N(z)$ estimates. Also created an independent photo-z validation set.

DP1 Photo-z - Created an independent photo-z validation set with DESI cross matched spec-z to test and validate DP1 photo-z estimates.

Active Optics - Helped characterize active optics system by studying variation in measured and fit parameters between nights.

SENSEI

SENSEI backgrounds - Characterized backgrounds for SENSEI dark matter direct detection experiment. Optimized masks and binning parameters to maximize signal on Skipper CCDs.

BNL

Parity Violation - Worked on detecting parity violation signatures in Lyman-α data. Found strong spurious signal that would come from standard covariance estimates and developed a more conservative covariance matrix that produced a null signal.

Generalized Redundant Calibration - Developed an improvement to redundant calibration for radio telescope arrays. Showed that while possible, including model data mismatches made it a poor fit.

Honors and Awards

DOE SCGSR Fellowship ($10,800) — Department of Energy - 2025
Lourie Fellowship ($6,000) — Stony Brook University - 2024
Gerald Brown Prize ($1,000) — Stony Brook University - 2023
DOE SULI ($12,000) — Department of Energy - 2020
DOE SULI ($5,300) — Department of Energy - 2018

Publications

Major Author

[1] Prakruth Adari, Xiangchong Li, et al. "Improvements to Shear Calibration Bias via Removal of Blends from Different Redshifts". In: (2026, in prep.).

[2] Anja von der Linden, Prakruth Adari, et al. "A Weak-Lensing View of Abell 360 in Vera Rubin Observatory's Data Preview 1". In: (2026, in prep.).

[3] Shuang Liang, Prakruth Adari, et al. "Catalog-based detection of unrecognized blends in deep optical ground based imaging". In: *The Open Journal of Astrophysics* 9 (May 2026), p. 62504. DOI: 10.33232/001c.162504. arXiv: 2503.16680 [astro-ph.CO].

[4] Prakruth Adari and Anja von der Linden. *SITCOMTN-128: Unrecognized Blends in LSSTComCam Data Preview 1 ECDFS*. Jan. 2025. DOI: 10.71929/RUBIN/2570850.

[5] Prakruth Adari and Anže Slosar. "Searching for parity violation in SDSS DR16 Lyman-α forest data". In: *Phys. Rev. D* 110.10 (2024), p. 103534. DOI: 10.1103/PhysRevD.110.103534. arXiv: 2405.04660 [astro-ph.CO].

[6] Prakruth Adari and Anže Slosar. "Generalized redundant calibration of radio interferometers". In: *Phys. Rev. D* 106.4 (2022), p. 043006. DOI: 10.1103/PhysRevD.106.043006. arXiv: 2107.10186 [astro-ph.IM].

Contributing Author

[1] Conghao Zhou, Xiangchong Li, et al. "Accurate Galaxy Cluster Shear and Mass Calibration for LSST with AnaCal". In: *arXiv e-prints*, arXiv:2606.21141 (June 2026), arXiv:2606.21141. DOI: 10.48550/arXiv.2606.21141. arXiv: 2606.21141 [astro-ph.CO].

[2] T. Zhang, E. Charles, et al. "Photometric Redshift Estimation for Rubin Observatory Data Preview 1 with Redshift Assessment Infrastructure Layers (RAIL)". In: *arXiv e-prints*, arXiv:2510.07370 (Oct. 2025), arXiv:2510.07370. DOI: 10.48550/arXiv.2510.07370. arXiv: 2510.07370 [astro-ph.IM].

[3] Ismael Mendoza et al. "The Blending ToolKit: A simulation framework for evaluation of galaxy detection and deblending". In: *Open J. Astrophys.* 8 (2025), p. 001c.129699. DOI: 10.33232/001c.129699. arXiv: 2409.06986 [astro-ph.IM].

[4] Eric Charles, John Franklin Crenshaw, et al. *SITCOMTN-154: Initial studies of photometric redshifts with LSSTComCam from DP1*. Jan. 2025. DOI: 10.71929/RUBIN/2571480.

[5] Celine Combet, Andreas Plazas Malagón, et al. *SITCOMTN-161: PSF assessment in the field of Abell 360 and shapeHSM shear profile using LSSTComCam data*. Jan. 2025.

[6] Miranda R. Gorsuch, Arun Kannawadi, et al. *Testing the implementation of Metadetection and Cell-Based Coadds on Abell 360 LSSTComCam data*. Jan. 2025.

[7] NSF-DOE Vera C. Rubin Observatory Team, Tatiana Acero-Cuellar, et al. *RTN-095: The Vera C. Rubin Observatory Data Preview 1*. Jan. 2025. DOI: 10.71929/RUBIN/2570536.

[8] Prakruth Adari et al. "First Direct-Detection Results on Sub-GeV Dark Matter Using the SENSEI Detector at SNOLAB". In: *Phys. Rev. Lett.* 134.1 (2025), p. 011804. DOI: 10.1103/PhysRevLett.134.011804. arXiv: 2312.13342 [astro-ph.CO].

[9] Prakruth Adari, Roman Berens, et al. "Charging up boosted black holes". In: *Phys. Rev. D* 107.4 (2023), p. 044055. DOI: 10.1103/PhysRevD.107.044055. arXiv: 2111.15027 [gr-qc].

[10] P. Adari, A. Aguilar-Arevalo, et al. "EXCESS workshop: Descriptions of rising low-energy spectra". In: *arXiv e-prints*, arXiv:2202.05097 (Feb. 2022), arXiv:2202.05097. DOI: 10.48550/arXiv.2202.05097. arXiv: 2202.05097 [astro-ph.IM].

Talks and Posters

Invited Talks

Seminar, AAS 247	Phoenix, Arizona - 2026
Seminar, ASIAA	Taipei, Taiwan - 2025
Talk, Parity Violation from Home	Online - 2024
Seminar, DESI Lyman-α Working Group	Online - 2024

Contributed Talks

Plenary, DESC Collaboration Meeting	Urbana-Champaign, IL - 2025
Plenary, DESC Collaboration Meeting	Menlo Park, CA - 2023

Posters

DESC Collaboration Meeting	Urbana-Champaign, IL - 2025
Cosmology and galaxy astrophysics with simulations and machine learning	New York, NY - 2024
DESC Collaboration Meeting	Zürich, Switzerland - 2024
DESC Collaboration Meeting	Menlo Park, CA - 2023

Teaching Experience

Teaching Assistant

ASTRO 443: Observational Techniques in Astronomy	2023
PHY 134: Classical Physics Laboratory II	2023
PHY 121: Physics for the Life Sciences I	2022
IACS Python Summer Bootcamp	2021

Mentoring

Gunjan Deshpande

Graduate student in Data Science at Stony Brook. Worked on investigating possible improvements to catalog-based detection of unrecognized blends by including AGNs and testing Random Forest variants (XGBoost, Extremely Randomized Forests).

Sheina Peralta

Undergraduate student in Astronomy at Stony Brook. Worked on extending previous catalog level machine learning techniques to include detecting AGNs and strong lenses along with

investigating the radial number density of red-sequence cluster members in Abell 360 analysis.

Institutional Service

Quality of Life Analysis

Ran a "Quality of Life" survey to assess students mental and physical wellbeing along with collecting views on the department. Helped design, disseminate, and analyze survey. Culminated into a presentation for the faculty and chair of the department leading to various improvements including pay and increased academic support for incoming students.

PGSA Treasurer

Served as Physics Graduate Student Association (PGSA) treasurer from 2023-2025. Helped schedule and run tea time, organize and procure materials for social events, and schedule practice talks during the semester.